v1.0Magneton Deep Dive_ Benevolent Monopole

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Magneton Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026
Applies to: `v1.0Magneton Deep Dive_ Benevolent Monopole V2.0.md`

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines' Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

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V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
MoEDAL-MAPP Run 3	Pre-Run 3 search	MoEDAL+MAPP 2023: σ < 0.5 fb for m_M > 100 GeV. No signal. Mass floor updated	MEDIUM	Exclusion limit tightened; simulation parameter space constrained
Electroweak monopole	Only Dirac monopole covered	Cho-Maison monopole: mass ~4-10 TeV, LHC-accessible. V2.0: EW_MONOPOLE type with m=7 TeV	HIGH	Lighter, testable monopole category entirely absent from V1.0
Condensed matter analogues	Spin ice mentioned	Spin ice monopoles ($Ho_2Ti_2O_7$): deconfinement at T_D~0.6 K confirmed. CMS analogue mode added	MEDIUM	Physically realized monopole system not modeled in V1.0
Dirac charge value	g_D = n*hbar*c/(2e)	g_D = 68.5×e per Dirac unit at n=1. V2.0: interaction force table updated with 68.5^2 factor	LOW	Precise charge value needed for correct force calculation

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Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** Electroweak
- **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
- **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
- **New interaction modes:** Triggered by new experimental discoveries

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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Magneton Deep Dive_ Benevolent Monopole V1.0 archived | V2.0 is the active specification as of June 2026